

Finsim AI

AI-Powered Investment Learning and Market Simulator

Author: Aravind, B.Tech CSE (AI & ML), Siddhartha Institute of Technology & Sciences

1. What This Project Is

Finsim AI is a web platform where users **practice investing with virtual money** while an AI explains real market behavior in plain English. It is explicitly an **education and simulation tool** — not a real trading system, not investment advice, and not a SEBI-regulated advisory product.

One-line description:

'An AI that watches real stock market data, teaches concepts using real examples, and lets users practice buying/selling with virtual money — so they understand investing before ever risking real money.'

2. Why This Name and Framing

Old Framing	Current Framing
'AI broker agent' that recommends real trades	'Simulator' for learning
Legal gray zone (advice-adjacent, SEBI RIA concerns)	Clearly educational — no licensing required
Real trading focus	Practice + understanding focus

Positioning as a **learning simulator** (not a trading bot) keeps this legally clean and makes it a strong academic project as well as a genuine product idea. The word 'SIMULATION' or 'Virtual' stays visible directly in the UI on every relevant screen — not buried in documentation — so the legal framing holds up in practice, not just on paper.

3. Core User Flow

- User signs up** → gets ■1,00,000 / \$100,000 virtual cash.
- Platform pulls real market data** (yfinance) — never synthetic or randomly generated prices, so patterns learned actually reflect real market behavior.
- AI analyzes each stock** and explains signals in plain English, grounded in that stock's actual computed numbers (never generic templates).
- User practices buying/selling with virtual money**, with real brokerage, STT, and capital-gains tax deducted on every trade — so the numbers feel real.
- AI teaches the 'why'** behind each concept as it comes up, optionally read aloud via voice narration.
- User sees a report of virtual profit/loss**, mistakes, and lessons learned over time, alongside volatility/drawdown — not just a smoothed return figure.

4. Feature List

4.1 Core Features

- **Virtual portfolio:** Starting balance (■1,00,000 / \$100,000), buy/sell paper simulation with live P&L.;
- **Real market data:** yfinance with 2-second streaming — includes real market-regime periods (2020 crash, 2021 rally, 2022 dip, 2023-2025 bull run, 2026 live) so users experience genuine volatility.
- **Up-to-Date Financial Overview:** 69 months of institutional closing data (2021–2025) with dynamic real-time live streaming for 2026.
- **Technical indicator engine:** Wilder's 14-period RSI, EMA-12/26 MACD, 20-period 2-std dev Bollinger Bands, Volume ratio.
- **AI-generated plain English explanations:** Grounded in that stock's real computed numbers.
- **Daily virtual portfolio report:** Mini 'EOD summary' showing volatility/drawdown alongside returns.

4.2 Education-Focused Features

- **Concept lessons tied to real signals:** Concept lessons explain technical indicators the moment a stock triggers oversold or golden crosses.
- **Voice narration:** Browser's free built-in SpeechSynthesis API (no API key, zero cost).
- **Historical win-rate shown per signal type:** E.g. 'RSI < 30 has preceded a bounce ~60% of the time in backtests' — keeps every AI explanation honest.
- **Realistic costs simulation:** Brokerage + STT deducted on every virtual trade; short-term capital gains tax shown in reports.
- **Diversification score/warning:** Alerts user if virtual portfolio is over-concentrated in one stock or sector.
- **Market regime replay mode:** Real historical periods, not smoothed averages.

4.3 Engagement / Behavior Features

- **Loss 'streak' mechanic:** Consecutive losses trigger a short mandatory cooldown before the next trade, giving virtual losses behavioral weight.
- **Mandatory reflection prompt:** Required reflection after a losing trade before the next action is allowed.
- **Progress tracking:** Virtual P&L; history, lessons completed, mistakes flagged over time.

4.4 Data & Scale Approach

- **Starter watchlist:** ~20 liquid, well-known NSE stocks across sectors for the initial build.
- **Scaled up to full market coverage:** 2,400+ securities (all 2,298+ active Indian NSE stocks in  plus 100+ Global US market leaders in \$).
- **AI explanations cached:** Cached per stock per day and served to all users watching that stock, staying comfortably within free-tier LLM limits.

4.5 Implemented & Enhanced Architecture

- **Full NSE market screener & catalog:** 2,400+ securities with instant search, sector tabs, and 2-second streaming ticks.
- **Multi-user authentication:** Mandatory Full Name validation on sign-up before OTP or Google verification.
- **Interactive Alerts & Signals Drawer:** Real unread count tracking, category badges (AI SIGNAL, TRADE FILL, SYSTEM), and filter tabs.
- **Complete Dark / Light Mode:** Dynamic chart strokes and Tailwind CSS v4 custom variant.
- **Optional real-broker paper trading connection:** Zerodha Kite Personal API protocol (free tier) for advanced users, strictly human-approved only.

5. Tech Stack

Layer	Technology	Cost
Frontend Hub	Next.js 16.3.4 (React 19), Tailwind CSS v4, Lucide React, Recharts	Free / Open Source
Backend / Tech Engine	Python 3.10+, Flask, Pandas, NumPy	Free / Open Source
Database	SQLite3 (investiq.db, finsim.db) → PostgreSQL (production)	Free
Stock Data	yfinance (sub-2s streaming in-memory cache)	Free
Full NSE Stock List	NSE official equity list CSV (2,298+ equities) + Global US catalog	Free
AI Explanations	Groq LLM (Llama 3.1) / OpenAI-compatible API	Free tier
Voice Narration	Browser SpeechSynthesis API	Free, built-in
Auth	Flask-Login & Next.js session cookies (with mandatory Full Name)	Free
Hosting	Render / Railway (backend), Vercel (frontend)	Free tier
Scheduler	GitHub Actions / Background daemons	Free

Total cost to build and run at small scale: 

6. Project File Structure

```
Invest_iq/
■■■■ app.py # Flask app & API reverse-proxy gateway
■■■■ data_provider.py # yfinance data fetching, live/mock toggle
■■■■ scanner.py # Runs technical analysis across watchlist/market
■■■■ indicators.py # RSI, MACD, Bollinger, volume ratio calculations
■■■■ explainer.py # Builds plain-English explanations from real numbers
■■■■ signal_stats.py # Tracks and returns historical win-rate per signal type
■■■■ portfolio.py # Virtual portfolio logic – buy/sell, P&L, fees, tax
■■■■ auth.py # User accounts and sessions
■■■■ models.py # Database schema (users, trades, watchlists, signal history)
■■■■ broker_kite.py # Zerodha Kite Connect API paper bridge
■■■■ build_history.py # 2021-2026 multi-year benchmark history extractor
■■■■ all_monthly_history.json# 69-month institutional closing data (Nifty 50 & S&P; 500)
■■■■ generate_pdf.py # Automated Project Overview PDF generator
■■■■ Invest_IQ_Project_Overview.pdf # Generated PDF Project Overview
■
■■■■ frontend/ # Next.js 16 (React 19) Full-Stack Web Application
■ ■■■■ src/app/ # App Router (Dashboard, Markets, Trade, Learn, Auth)
■ ■■■■ src/components/ # Topbar (Bell+Theme), Sidebar, NotificationsDrawer, PortfolioChart
■ ■■■■ src/lib/ # financial-history.ts, market-api.ts, store.tsx, catalogs
■
■■■■ templates/ # Flask Jinja2 templates (dashboard, screener, portfolio, report)
■■■■ static/ # CSS & static assets for Flask engine
■■■■ tests/ # Automated verification test suite
■■■■ requirements.txt # Python package dependencies
```

7. Build Roadmap & Status

Phase	Goal	Status
1	Data pipeline — fetch real historical stock data via yfinance with retry handling	■ Completed
2	Technical indicators — RSI, MACD, Bollinger Bands, volume ratio computed and stored	■ Completed
3	Signal detection + plain English explainer grounded in real numbers with visible disclaimer	■ Completed
4	Virtual portfolio — buy/sell simulation with real brokerage, STT, and capital-gains tax deducted per trade	■ Completed
5	Historical win-rate tracking per signal type, surfaced transparently next to every AI explanation	■ Completed
6	Behavioral guardrails — loss-streak cooldown and mandatory reflection prompt after a losing trade	■ Completed
7	Voice narration (browser SpeechSynthesis API) on explanations and reports	■ Completed
8	Authentication — real user accounts with mandatory Full Name validation on sign-up	■ Completed
9	Scale stock list to full market — 2,400+ securities (2,298+ NSE + 100+ Global mega-caps) with 2s live streaming	■ Completed
10	Up-to-date Financial Overview — 69 months real institutional close data (2021–2025) and dynamic live 2026 streaming	■ Completed
11	Interactive alerts & dark/light mode — slide-over drawer with unread tracking and Tailwind v4 theme sync	■ Completed
12	Cloud deployment configuration (Docker, Render, Vercel, Railway) ready for staging & production	■ Completed

8. Legal Positioning

- Finsim AI is always framed as:
- A **learning and simulation tool**, not investment advice.
 - Virtual money only in the core product — any future real-money connection remains fully human-approved, with no autonomous trading.
 - Every AI-generated signal explanation carries a clear disclaimer and its real historical win-rate — never a confident directive to 'buy' or 'sell'.
 - The word '**SIMULATION**' or '**Virtual**' visible directly in the UI on every relevant screen.

This framing keeps the project outside SEBI Registered Investment Adviser (RIA) requirements, which apply to personalized real-money investment advice, not education/simulation tools.